MSE 2 CASE STUDY REPORT

E-Waste Management, Circular Economy & Recycling Models

Based on the 10 case studies listed in the provided TE case-study sheet

Student Name: _____

Roll No.: _____ Class/Division: _____

Subject: _____

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Source Basis and Scope

The uploaded TE case-study sheet lists ten topics: Japan recycling, Switzerland e-waste, India's E-Waste Management Rules 2022, India's EPR, Bengaluru, Mumbai, Clean Kerala Company, Pune SWaCH, Dharavi and Dell's circular-economy strategy. ■filecite■turn0file0■L2-L12■

This report expands those titles into exam-ready case studies using official government/organizational information and research sources. Where city-level figures are old or methodology-specific, they are explicitly labelled as historical estimates rather than presented as current official totals.

Important distinction: some case studies concern municipal solid waste or plastic recycling rather than e-waste alone. They are included because the requested case-study list uses them to illustrate broader recycling, collection, informal-sector and circular-economy lessons relevant to e-waste management.

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1. Why Japan Recycles More Than Most Countries: Lessons for India

Executive Summary

Japan demonstrates a mature waste-management system built around waste reduction, source separation, municipal responsibility, extensive treatment infrastructure and public participation. The important lesson for India is not simply to copy Japan, but to adapt its discipline of segregation, predictable collection, data-based planning and producer responsibility to Indian urban conditions.

Background

Japan has pursued waste reduction and recycling for decades through national policy, municipal systems and public participation. The Ministry of the Environment reported 38.11 million tonnes of municipal waste in FY2024, or 839 g per person per day. Its recycling rate was 19.3%, while 99.2% of waste was reduced through recycling and/or intermediate treatment before final disposal. Japan also had 991 municipal incineration plants as of March 31, 2025, with 41.9% equipped for power generation.

Strategy / Initiative

- Strict separation and collection rules at municipal level.
- Waste reduction and reuse before disposal.
- High-capacity intermediate treatment and controlled incineration.
- Public participation through clear local sorting instructions and schedules.
- Continuous measurement of generation, treatment, recycling and landfill indicators.

Impact & Results with Data

Indicator	Japan FY2024	Meaning
Municipal waste generated	38.11 million t	2.2% lower than FY2023
Waste per person/day	839 g	1.5% lower than FY2023
Recycling rate	19.3%	7.38 million t recycled
Waste reduction rate	99.2%	Very little untreated waste reaches landfill
Incineration plants	991	41.9% had power generation facilities

SWOT Analysis

Strengths	Weaknesses
Strong municipal discipline; clear separation; extensive infrastructure; reliable data	High capital and operating costs; incineration remains important; system is difficult to reproduce exactly in dense, informal settings
Opportunities	Threats
Adapt segregation, deposit/take-back ideas and data systems in Indian cities	Population growth, changing consumption patterns and contamination of recyclables can reduce performance

Comparison with India

India has a much larger and more heterogeneous waste sector, with substantial informal recycling and uneven source segregation. Japan’s main transferable lesson is system discipline: make segregation easy, collect it reliably, measure outcomes and enforce responsibility across the chain. India should combine this with the strengths of its informal waste workers rather than displacing them.

Recommendations

- Adopt city-level standardized dry/wet/e-waste segregation instructions.

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- Publish ward-level collection and recycling dashboards.
- Integrate informal collectors into formal systems through IDs, training and safe collection points.
- Use producer take-back for electronics and packaging where appropriate.
- Measure landfill diversion and material recovery separately.

Conclusion

Japan shows that recycling performance depends on an entire system rather than a single recycling plant. For India, the most useful adaptation is consistent segregation, collection, data reporting and accountability, supported by infrastructure and citizen participation.

References

- Ministry of the Environment, Japan — Municipal solid waste generation and disposal in FY2024 (2026).
- Ministry of the Environment, Japan — Municipal solid waste generation and disposal in FY2023 (2025).

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2. Switzerland's E-Waste Success Story

Executive Summary

Switzerland's e-waste system is based on producer/importer responsibility, advance recycling financing, convenient take-back and specialized recycling organizations. SENS eRecycling and SWICO have operated take-back systems for decades, creating a stable channel from consumers to professional recyclers.

Background

Switzerland uses established take-back systems for electrical and electronic equipment. In 2023, SENS and SWICO recycling companies processed about 132,000 tonnes of waste electrical and electronic equipment, 9% more than in 2022. SENS alone reported 92,874 tonnes processed in its system and 509 official collection points in 2023.

Strategy / Initiative

- Convenient return through collection points and retail channels.
- Advance financing/fees support the recycling system.
- Specialized producer/importer organizations coordinate take-back.
- Professional recycling separates recoverable materials and hazardous components.
- Transparent annual reporting builds accountability.

Impact & Results with Data

Indicator	2023	Observation
SENS + SWICO processed	~132,000 t	9% increase from 2022
SENS system	92,874 t	10.36 kg per resident
Official SENS collection points	509	Stable nationwide network
Manufacturers/importers subject to SENS ARF	1,381	7% increase from 2022

SWOT Analysis

Strengths	Weaknesses
Convenient collection; stable financing; specialized recyclers; reporting	Requires mature institutions and financing; performance data depend on system boundaries
Opportunities	Threats
Higher repair/reuse and circular design; stronger digital tracking	Rapid device turnover, complex products and batteries create new treatment challenges

Comparison with India

India's E-Waste Management Rules, 2022 also use Extended Producer Responsibility and a CPCB portal, but India's scale and informal sector make implementation more complex. Switzerland demonstrates the value of convenient take-back and predictable financing. India can adapt these mechanisms while building formal links with existing collectors and refurbishers.

Recommendations

- Increase easily accessible authorized collection points in Indian cities.
- Make producer take-back information visible at point of sale.
- Strengthen EPR certificate verification and audit.
- Create safe battery and small-device collection streams.

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• Publish recycler performance indicators in a consumer-friendly format.

Conclusion

Switzerland's experience shows that e-waste recycling becomes easier to scale when financing, collection and recycling responsibilities are clearly assigned and convenient for consumers.

References

• SENS eRecycling — Facts & Figures 2023.

• Swiss Recycle — 132,000 tonnes processed by SENS and SWICO in 2023.

• SENS/SWICO Technical Report and annual reporting.

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3. India's E-Waste Management Rules 2022

Executive Summary

The E-Waste (Management) Rules, 2022 represent India's shift toward a stronger Extended Producer Responsibility framework. The rules came into force on 1 April 2023 and use a centralized EPR portal, recycling obligations, EPR certificates, refurbishing certificates, environmental compensation and audit provisions.

Background

CPCB states that the 2022 Rules cover 106 electrical and electronic equipment items under seven categories. Producers meet EPR obligations through registered recyclers and the EPR portal. CPCB's FAQ identifies recycling targets, EPR certificates, refurbishing certificates, environmental compensation, returns and audits as important features.

Strategy / Initiative

- Register producers, manufacturers, refurbishers and recyclers on the CPCB system.
- Set producer recycling obligations and allow fulfillment through EPR certificates.
- Use registered recyclers for environmentally sound recycling.
- Encourage reuse/refurbishment through refurbishing certificates.
- Use environmental compensation, returns and audits for compliance.

Impact & Results with Data

Indicator	Documented position	Significance
Rules effective	1 Apr 2023	Current regulatory framework
EEE covered	106 items / 7 categories	Broader scope than earlier framework
EPR certificates	Generated through portal	Market mechanism for obligation fulfillment
Refurbishing certificates	Provided	Encourages life extension
Environmental compensation	Provided	Penalty/compliance mechanism

SWOT Analysis

Strengths	Weaknesses
National framework; centralized portal; producer responsibility; audit and compensation provisions	Implementation depends on accurate reporting, verification and adequate formal recycling capacity
Opportunities	Threats
Formalization of informal sector; better tracking; repair/reuse growth	False reporting, weak traceability, low consumer participation and capacity gaps

Comparison with India

Compared with Switzerland, India's framework has similar producer-responsibility logic but operates at a far larger scale and in a market where informal collection is substantial. Compared with Japan, India's key challenge is less about basic municipal waste discipline and more about traceability, formal channelization and compliance across a complex market.

Recommendations

- Strengthen third-party verification of recycler output.
- Improve consumer awareness of authorized take-back options.
- Create formal pathways for informal collectors to supply registered recyclers.
- Track material recovery and reuse separately from simple collection.

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• Publish periodic state-wise compliance dashboards.

Conclusion

The 2022 Rules provide a stronger policy architecture for e-waste management. Their long-term effectiveness depends on verification, enforcement, consumer participation and integration of existing collection networks.

References

• CPCB — E-Waste (Management) Rules, 2022 FAQ.

• MoEFCC — E-Waste (Management) Rules, 2022 and amendments.

• CPCB E-Waste Management System / EPR Portal.

• PIB, Ministry of Environment — Generation of E-waste, 27 July 2023.

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4. India's Extended Producer Responsibility (EPR): Success or Paper Compliance?

Executive Summary

EPR shifts end-of-life responsibility toward producers rather than leaving waste management entirely to municipalities and consumers. In India, EPR under the 2022 e-waste rules creates measurable obligations, but the distinction between formal compliance and actual environmentally sound recycling is crucial.

Background

India's earlier framework used EPR authorization and collection targets. The 2022 rules moved toward recycling-based obligations, EPR certificates and a centralized portal. CPCB's FAQ states that producers must fulfill EPR obligations through registered recyclers and that audits and environmental compensation are part of the framework.

Strategy / Initiative

- Producer registration and obligation calculation.
- Purchase/transfer of EPR certificates from registered recyclers.
- Recycler registration and reporting.
- Audits and environmental compensation for non-compliance.
- Refurbishment pathway to extend product life.

Impact & Results with Data

Measure	Evidence available	Interpretation
EPR portal	National system operational	Creates a common reporting channel
Producer obligation	Defined under Rules/Schedules	Makes responsibility measurable
Recycler linkage	Registered recyclers required	Reduces reliance on unverified recycling
Audits	Provided in rules	Important for verification
Environmental compensation	Provided	Creates compliance consequence

SWOT Analysis

Strengths	Weaknesses
Clear legal responsibility; digital portal; certificates; audit provisions	Reported recycling may not always equal independently verified material recovery; enforcement is complex
Opportunities	Threats
Digital traceability; stronger audits; integration with informal sector	Certificate misuse, data quality problems and low-quality downstream processing

Comparison with India

Switzerland has long-established take-back financing and collection networks; India has a newer centralized digital EPR mechanism. India's challenge is therefore not only policy design but verification at the physical end of the chain. A useful benchmark is whether reported EPR activity results in traceable, environmentally sound material recovery.

Recommendations

- Use risk-based physical audits of recyclers.
- Link EPR certificates to batch-level material recovery evidence where feasible.
- Publish producer compliance rates and recycler performance.

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• Recognize verified collection partnerships with informal aggregators.

• Separate reuse/refurbishment outcomes from recycling outcomes.

Conclusion

EPR should be judged by traceable environmental outcomes, not only registrations or certificates. India's 2022 framework contains the mechanisms needed for this, but implementation quality determines whether EPR becomes substantive or mainly administrative.

References

• CPCB — E-Waste (Management) Rules, 2022 FAQ.

• CPCB — E-Waste Management System.

• CPCB — EPR fact sheet and earlier EPR framework.

• MoEFCC — E-Waste Rules and amendments.

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5. Bengaluru—India's E-Waste Capital?

Executive Summary

Bengaluru's technology-intensive economy has made it an important Indian e-waste case study. Historical studies identified a large flow of obsolete computers and electronics, while the city's broader waste system illustrates the need to connect consumers, repair shops, traders, dismantlers and formal recyclers.

Background

Older field studies estimated Bengaluru generated around 12,000 tonnes of e-waste per year from computers and peripherals. Another academic review reported that tens of thousands of computers became obsolete annually in the IT sector. These figures are historical estimates and should not be treated as current city totals.

Strategy / Initiative

- Formal recycling companies and collection channels.
- Repair, reuse and secondary markets for electronics.
- Informal traders and dismantlers recover materials.
- Awareness and institutional collection can channel waste toward authorized recyclers.
- EPR provides a national framework for producer responsibility.

Impact & Results with Data

Data point	Reported figure	Caution
Historical e-waste estimate	~12,000 t/year	Computer/peripheral estimate from older field study
Historical IT obsolescence	30,000–40,000 computers/year	Older academic estimate
City role	Major technology/IT hub	Supports high electronics turnover

SWOT Analysis

Strengths	Weaknesses
Large technology ecosystem; repair/reuse culture; presence of formal recyclers	Historical data are inconsistent; informal handling can be difficult to trace; hazardous dismantling risks
Opportunities	Threats
IT-company take-back; refurbishment; campus collection; formal-informal partnerships	Rapid device replacement, informal unsafe processing and data-security concerns

Comparison with India

Bengaluru's situation differs from Japan or Switzerland because a large informal market can coexist with formal recyclers. India can use Bengaluru as a pilot for digital take-back, corporate e-waste collection and integration of repair/refurbishment with registered recycling.

Recommendations

- Require large institutions to document e-waste handover to authorized recyclers.
- Create campus and IT-park collection drives.
- Support certified refurbishment before recycling.
- Provide safe dismantling training and PPE for workers moving into formal channels.
- Use city-level e-waste dashboards instead of relying on old estimates.

Conclusion

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Bengaluru is a useful case because technology growth creates both an e-waste challenge and an ecosystem capable of solving it. Current planning should rely on updated measurements rather than historical estimates.

References

• E-Waste Management: A Case Study of Bangalore, India — field study.

• Electronic waste – an emerging threat to the environment of urban India — academic review.

• CPCB/PIB — national e-waste framework.

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6. Mumbai's E-Waste Ecosystem

Executive Summary

Mumbai combines a very large consumer and commercial base with a long-established informal recycling economy. The case highlights the importance of collection networks, scrap markets, municipal systems and formal recycling capacity, while also showing why city-level e-waste estimates can vary by methodology and year.

Background

A historical assessment estimated Greater Mumbai's annual e-waste at about 31,622 tonnes in 2012, while the wider Mumbai Metropolitan Region was estimated at about 47,469 tonnes. These are historical estimates and are included to illustrate scale, not as current official totals. Mumbai also has a large general solid-waste stream, making source segregation and channelization important.

Strategy / Initiative

- Household and institutional collection through scrap and recycling networks.
- Formal recyclers for environmentally sound processing.
- Municipal solid-waste infrastructure for dry-waste separation.
- Producer responsibility and authorized take-back under national e-waste rules.
- Potential integration of informal collectors into formal supply chains.

Impact & Results with Data

Historical indicator	Estimate	Use
Greater Mumbai e-waste	31,622 t/year (2012)	Historical estimate
Mumbai Metropolitan Region	47,469 t/year (2012)	Historical estimate
Current broader waste context	~6,200 t/day municipal solid waste in 2022–23 dataset	Not an e-waste figure

SWOT Analysis

Strengths	Weaknesses
Large recycling markets; dense collection network; strong commercial base	Informal processing risks; mixed-waste contamination; fragmented data
Opportunities	Threats
Formal-informal partnerships; authorized collection points; producer take-back	Unsafe dismantling, hazardous exposure, weak traceability and landfill pressure

Comparison with India

Mumbai's informal market can be an asset if linked to registered recyclers. Switzerland offers a model of convenient take-back and transparent financing; Mumbai can adapt those principles without eliminating local livelihoods. Japan's emphasis on segregation and municipal coordination is also relevant.

Recommendations

- Map e-waste collectors and aggregators by ward.
- Provide authorized drop-off points for households and small businesses.
- Build contracts between informal aggregators and registered recyclers.
- Conduct worker-safety and hazardous-material training.
- Publish updated city e-waste estimates using one transparent methodology.

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Conclusion

Mumbai demonstrates that e-waste management is an ecosystem problem. The objective should be to connect existing collection and recovery activity to safe, traceable and formally accountable recycling.

References

• Historical Mumbai e-waste assessment / academic literature.

• Municipal and metropolitan waste datasets cited for broader waste context.

• CPCB E-Waste Management Rules, 2022 FAQ.

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7. Kerala's Clean Kerala Company

Executive Summary

Clean Kerala Company Limited (CKCL), under Kerala's Local Self Government Department, provides forward linkages for local governments in waste processing and disposal. Its e-waste model connects local collection, Haritha Karma Sena and authorized recycling companies.

Background

CKCL is a government undertaking that works with local governments, Haritha Karma Sena, LSGD and other partners. The Kerala government reports a broad doorstep non-biodegradable collection network. CKCL states that it collects e-waste and hands it to e-waste recycling companies approved by the relevant pollution-control authorities.

Strategy / Initiative

- Doorstep collection through local-government-linked systems.
- Haritha Karma Sena as a collection workforce.
- Material collection and aggregation by CKCL.
- Handover of e-waste to approved recycling companies.
- Separate handling arrangements for hazardous e-waste.

Impact & Results with Data

Indicator	Reported figure	Source period/context
CKCL e-waste collected	2,678 MT	Current CKCL website figure
Hazardous e-waste	205 MT	Current CKCL website figure
LSGD/Haritha Karma Sena members	35,352	Up to May 2024
Households with non-biodegradable doorstep collection	7,711,000	Up to May 2024 cumulative figure

SWOT Analysis

Strengths	Weaknesses
Government-backed network; doorstep collection; integration with local bodies; separate hazardous stream	Dependence on local implementation and downstream recycler capacity
Opportunities	Threats
Expand electronics collection campaigns; digital tracking; repair/refurbishment	Growing e-waste, hazardous components and contamination of collected material

Comparison with India

Kerala provides a useful Indian example of decentralized collection feeding into formal downstream recycling. Compared with Switzerland, the financing and institutional context differ, but both show the value of convenient collection. Compared with Japan, Kerala's strength is the linkage between local community collection and a state-level coordinating company.

Recommendations

- Give CKCL-style aggregation support to other states.
- Add QR-based collection receipts and downstream tracking.
- Create special drives for phones, computers, appliances and batteries.

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- Measure reuse/refurbishment separately from recycling.
- Publish district-level e-waste collection and downstream treatment data.

Conclusion

Clean Kerala Company demonstrates how a state-level coordinating organization can connect local collection systems to formal recycling. Its main lesson is institutional coordination across local governments, collection workers and authorized recyclers.

References

- Clean Kerala Company Limited official website — Activities at a Glance.
- Kerala LSGD — CKCL overview.
- Haritham Keralam Mission — waste management activity data.

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8. Pune's SWaCH Model

Executive Summary

SWaCH is a cooperative model integrating waste pickers into Pune's solid-waste management system. Although it is primarily a municipal solid-waste case rather than an e-waste-only program, it provides an important model for recognizing informal workers as service providers and linking collection with recycling.

Background

SWaCH reports that by 31 December 2024 it had more than 3,900 integrated waste pickers, covered about 10 lakh properties and serviced around 40 lakh people. It reports approximately 1,500–1,600 tonnes of waste collected daily and about 220 tonnes per day diverted from landfill through recycling, with an estimated annual avoidance of 100,000 MT CO_2.

Strategy / Initiative

- Cooperative organization of waste pickers.
- Door-to-door collection linked with user interaction.
- Segregation and recovery of recyclable materials.
- Worker income and service delivery integrated into the system.
- Partnership with municipal solid-waste management.

Impact & Results with Data

Indicator	SWaCH reported impact	Context
Waste pickers integrated	3,900+	As of 31 Dec 2024
Properties covered	~10 lakh	Approximate
Population serviced	~40 lakh	Approximate
Waste collected	~1,500–1,600 t/day	SWaCH estimate
Waste diverted/recycled	~220 t/day	~35% diversion figure
Annual avoided emissions	~100,000 MT CO_2	SWaCH estimate

SWOT Analysis

Strengths	Weaknesses
Worker integration; large collection coverage; recycling diversion; cooperative structure	Model depends on local institutional support, user participation and stable recyclable markets
Opportunities	Threats
Extend the model to e-waste collection partnerships and repair networks	Commodity-price volatility, worker safety risks and policy changes

Comparison with India

India's e-waste system can learn from SWaCH's principle of integrating existing waste workers rather than treating them only as an informal problem. Switzerland provides a formal take-back benchmark; SWaCH offers an Indian social-infrastructure lesson: collection works better when workers and households are part of the system.

Recommendations

- Create an e-waste collection specialization within cooperative networks.
- Train selected waste pickers in safe identification and storage of electronics.
- Pay collection incentives for verified handover to registered recyclers.

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- Provide PPE and insurance/social protection mechanisms.
- Track collection quantities without mixing e-waste with general recyclables.

Conclusion

SWaCH shows that waste management can combine environmental outcomes with livelihood integration. Its approach is especially relevant to India's e-waste challenge because collection is often the missing link between consumers and formal recyclers.

References

- SWaCH Cooperative — Impact page, data through 31 Dec 2024.
- Pune municipal waste-management context.

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9. Dharavi Recycling Ecosystem

Executive Summary

Dharavi is widely discussed as an informal recycling and trading hub. Its importance lies in the scale and specialization of small enterprises that sort, trade and recover materials. The case also raises questions about worker safety, environmental controls, recognition and the relationship between informal and formal systems.

Background

Recent research on Compound 13 in Dharavi reports that the 13 Compounds area handles roughly 500–800 tonnes of plastic waste per day and functions as one of several recycling hubs in the Mumbai Metropolitan Region. Estimates cited in the research describe hundreds of recycling units and thousands of workers. These figures refer primarily to plastic recycling, not e-waste alone.

Strategy / Initiative

- Small-scale sorting and trading enterprises.
- Material specialization and recovery from discarded products.
- Dense buyer-seller networks that reduce transport and transaction costs.
- Reuse of recovered materials in secondary markets.
- Informal knowledge of material value and processing.

Impact & Results with Data

Indicator	Reported estimate	Important qualification
Plastic handled	~500–800 t/day	13 Compounds; primarily plastic
Annualized amount	~187,000 t/year	Research estimate; not an e-waste figure
Recycling units	~300–500	Reported estimate
Workers	~4,000–6,000	Reported estimate

SWOT Analysis

Strengths	Weaknesses
High material recovery; local expertise; employment; efficient trading networks	Informal conditions; occupational/environmental risks; limited official recognition and traceability
Opportunities	Threats
Formal partnerships, safer technology, worker recognition and data collection	Displacement without alternatives, stricter land-use pressure and unsafe processing

Comparison with India

Dharavi differs from Switzerland's regulated take-back system and Japan's municipal model, but it reveals an Indian advantage: dense informal recovery networks already exist. Formal policy can improve safety and traceability while preserving useful collection and market functions.

Recommendations

- Recognize aggregators and workers through simple registration.
- Provide safe storage, PPE and hazardous-material training.
- Create formal off-take agreements with registered recyclers.
- Collect reliable material-flow data before designing interventions.
- Avoid treating all informal recycling as identical; distinguish safe trading/reuse from hazardous processing.

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Conclusion

Dharavi demonstrates the scale of India's informal circular economy. The practical challenge is to improve safety, environmental performance and traceability while preserving legitimate livelihoods and material-recovery expertise.

References

• Springer research on Compound 13 Lab and Dharavi recycling ecosystem (2025).

• Research material on Compound 13 Lab / Dharavi recycling activity.

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10. Dell's Circular Economy Strategy

Executive Summary

Dell Technologies applies circular-economy principles across product design, repair, refurbishment, materials, packaging and end-of-life recovery. The strategy is a corporate case study showing how producer design decisions can reduce waste before products reach recycling.

Background

Dell describes six circularity principles including durability, repair/refurbishment, dematerialization, more sustainable materials, design for harvest and evolving business models. Dell reports that since 2007 it has recovered more than 2.5 billion pounds (about 1.1 billion kg) of used electronics. Its 2030 goals include reusing or recycling one metric ton for every metric ton of product purchased, 100% recycled/renewable/reused packaging, and more than half of product content from recycled, renewable or reduced-carbon materials.

Strategy / Initiative

- Design products for durability and easier repair/upgrade.
- Use recycled and renewable materials.
- Design components for disassembly and material recovery.
- Provide consumer and business recycling/take-back services.
- Refurbish usable products before recycling damaged products.
- Use closed-loop materials in selected product lines.

Impact & Results with Data

Indicator	Dell reported figure/goal	Meaning
Electronics recovered since 2007	>2.5 billion lb (~1.1 bn kg)	Historical cumulative recovery
2030 reuse/recycle goal	1 t for every 1 t purchased	Circular recovery ambition
2030 packaging goal	100% recycled/renewable/reused	Packaging circularity
2030 product-content goal	>50%	Recycled/renewable/reduced-carbon materials
Packaging reported on Dell India page	97.6%	100% recycled/renewable material, current webpage figure

SWOT Analysis

Strengths	Weaknesses
Lifecycle approach; repair/refurbishment; take-back; recycled materials; design-for-disassembly	Corporate goals depend on measurement, product design constraints and supply-chain execution
Opportunities	Threats
Longer product life, modular upgrades, more closed-loop materials and repair ecosystems	Rapid technology change, difficult component recovery and data-security concerns

Comparison with India

India's EPR framework and Dell's corporate circularity strategy address different levels of the system. EPR sets external responsibility; Dell demonstrates how a producer can act inside product design and recovery. India can encourage similar design-for-repair, refurbishment and material-recovery practices through standards, procurement and EPR incentives.

Recommendations

- Promote repairability and upgradeability in public procurement.

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- Reward verified reuse/refurbishment under circular-economy programs.
- Require clear end-of-life information and take-back options.
- Develop secure IT asset disposition for institutional electronics.
- Use recovered materials in new products where technically and economically feasible.

Conclusion

Dell's case shows that circular economy starts before disposal. Product design, repair, refurbishment, take-back and recycled materials can reduce the volume and environmental burden of e-waste reaching final recycling.

References

- Dell Technologies — Accelerating the Circular Economy.
- Dell — Circular Design Gives E-Waste a Second Life.
- Dell Technologies — Circular Economy / Design for Environment brief.
- Dell India — Sustainable IT and packaging information.

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One-Page Exam Revision Sheet

1. Japan: municipal discipline + source separation + treatment infrastructure + public participation. FY2024: 38.11 Mt waste, 839 g/person/day, 19.3% recycling rate.

2. Switzerland: producer/importer responsibility + convenient take-back + advance financing + specialized recyclers. 2023 SENS/SWICO: ~132,000 t processed.

3. India E-Waste Rules 2022: effective 1 Apr 2023; 106 EEE items under 7 categories; EPR certificates; refurbishing certificates; registered recyclers; audits and environmental compensation.

4. Indian EPR: success depends on physical outcomes and traceability, not only registration/certificates. Key issues: verification, reporting, recycler quality and informal-sector integration.

5. Bengaluru: technology hub + high electronics turnover + formal/informal recycling. Historical studies estimated ~12,000 t/year for computer/peripheral e-waste; use as historical evidence only.

6. Mumbai: large consumer/commercial base + informal scrap ecosystem + formal recyclers. Historical 2012 estimate: ~31,622 t/year Greater Mumbai e-waste; current city figures require updated methodology.

7. Clean Kerala Company: state-level coordinating company; local governments + Haritha Karma Sena + authorized recyclers. CKCL reports 2,678 MT e-waste and 205 MT hazardous e-waste collected on its current website.

8. SWaCH Pune: cooperative integration of waste pickers. By Dec 2024: 3,900+ workers, ~10 lakh properties, ~40 lakh people; ~1,500–1,600 t/day collected; ~220 t/day diverted/recycled.

9. Dharavi: informal recycling hub; research reports 13 Compounds handle ~500–800 t/day of mainly plastic waste, with hundreds of units and thousands of workers. Main lesson: integrate safety, recognition and traceability.

10. Dell: circular design + durability + repair/refurbishment + recycled materials + take-back + closed-loop recovery. Dell reports >2.5 billion lb used electronics recovered since 2007 and 2030 circularity goals.

Common exam conclusion: Effective e-waste management needs the full chain—design → reduce/reuse/repair → collection → safe transport → formal recycling → material recovery → monitoring. India can combine formal EPR with the strengths of its existing collection and informal-recycling networks.

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References Used for the Report

- Uploaded TE case-study list (provided source).
- Ministry of the Environment, Government of Japan — Municipal solid waste generation and disposal in FY2024.
- Ministry of the Environment, Government of Japan — Municipal solid waste generation and disposal in FY2023.
- SENS eRecycling — Facts & Figures 2023.
- Swiss Recycle / SENS / SWICO — 2023 e-waste processing report.
- Central Pollution Control Board — E-Waste (Management) Rules, 2022 FAQ and E-Waste Management System.
- Ministry of Environment, Forest and Climate Change — E-Waste (Management) Rules, 2022 and amendments.
- Press Information Bureau, Government of India — Generation of E-waste, 27 July 2023.
- International Telecommunication Union / UNITAR — Global E-waste Monitor 2024.
- Clean Kerala Company Limited — official activities and e-waste information.
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- SWaCH Cooperative — Impact data through 31 December 2024.
- Research literature on Bengaluru e-waste management and historical city estimates.
- Research literature on Mumbai e-waste ecosystem and historical estimates.
- Springer research on Compound 13 Lab / Dharavi recycling ecosystem.
- Dell Technologies — Accelerating the Circular Economy.
- Dell Technologies — Circular Design Gives E-Waste a Second Life.
- Dell Technologies — Circular Economy / Design for Environment.
- Dell India — Sustainable IT and packaging information.

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